20764 Administering a Microsoft SQL Database Infrastructure
Project 1

Introduction

This project will test your knowledge on the labs that you have completed in this course. You will work with the database you created in Assignment 5, making modifications as necessary. **Ensure that you have completed Assignments 5, 8 and 10 in their entirety** before you begin this project. In addition, before you begin, please ensure that you verify the information that is your RetailSales database, and re-run SQL commands as necessary.

Database Structure

The following tables should already represent the structure of your RetailSales database.

Database:	Retail Sales				
Table:	Ottawa.Customers		Table:	Ottawa.Suppliers	
Column Name	Character Type	Max Characters	Column Name	Character Type	Max Characters
CUSTOMERID	int (PK)		SUPPLIERID	int (PK)	
FIRSTNAME	varchar	100	SUPPLIERNAME	varchar	100
LASTNAME	varchar	100	ADDRESS	varchar	100
ADDRESS	varchar	100	CITY	varchar	100
CITY	varchar	100	PROVINCE	varchar	100
PROVINCE	varchar	100	POSTALCODE	varchar	100
PHONENUMBER	varchar	100	PHONENUMBER	varchar	100
Table:	Ottawa.Products		Table:	Transactions	
Column Name	Character Type	Max Characters	Column Name	Character Type	Max Characters
PRODUCTID	int (PK)		TRANSNUM	int (PK)	
DESCRIPTION	varchar	255	PRODUCTID	int (FK)	
COST	money		CUSTOMERID	int (FK)	
RETAILPRICE	money		RETAILPRICE	money	
SUPPLIERID	int (FK)		QUANTITY	int	

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Table Information:

Add the following information into the tables.

Customers Table			
CustomerID:	120	CustomerID:	121
FirstName:	Henriette	FirstName:	Rosina
LastName:	Samuel	LastName:	Strinati
Address:	120 Kings Cross Lane	Address:	121 Hopkins Lane
City:	Marionette	City:	Kingston
Province:	QC	Province:	ON
PostalCode:	R5T 6N8	PostalCode:	U8Y 9L1
PhoneNumber:	611-556-9852	PhoneNumber:	545-556-9866
CustomerID:	122	CustomerID:	123
FirstName:	Fillipo	FirstName:	Raffaele
LastName:	Candreva	LastName:	Spina
Address:	52 Ferrari Drive	Address:	12 Gianni De Luca St
City:	Folliero	City:	Vulcan
Province:	BC	Province:	AB
PostalCode:	Y9K 8K0	PostalCode:	P90 8T3
PhoneNumber:	151-456-7855	PhoneNumber:	611-556-9852
Suppliers Table			
SupplierID:	779562	SupplierID:	688559
SupplierName:	Multimicro Inc	SupplierName:	ChrisCo
Address:	29 Multimicro Way	Address:	73 Wild Goose Road
City:	Laval	City:	WaWa
Province:	QC	Province:	ON
PostalCode:	U8P 0Y7	PostalCode:	P0S 0A2
PhoneNumber:	819-966-3579	PhoneNumber:	552-669-9866
Products Table			
ProductID:	6689995	ProductID:	6685585
Description:	Logitech Ergo K860	Description:	High Speed 4K HDMI Cable, 25ft
Cost:	101.10	Cost:	12.00
RetailPrice	150.00	RetailPrice	19.99
SupplierID	779562	SupplierID	779562
Transactions Table			
TransNum:	55487766	TransNum:	382746
CustomerID:	122	CustomerID:	122
ProductID:	6689995	ProductID:	6689995
RetailPrice:	150.00	RetailPrice:	179.00
Quantity:	5	Quantity:	3

Query

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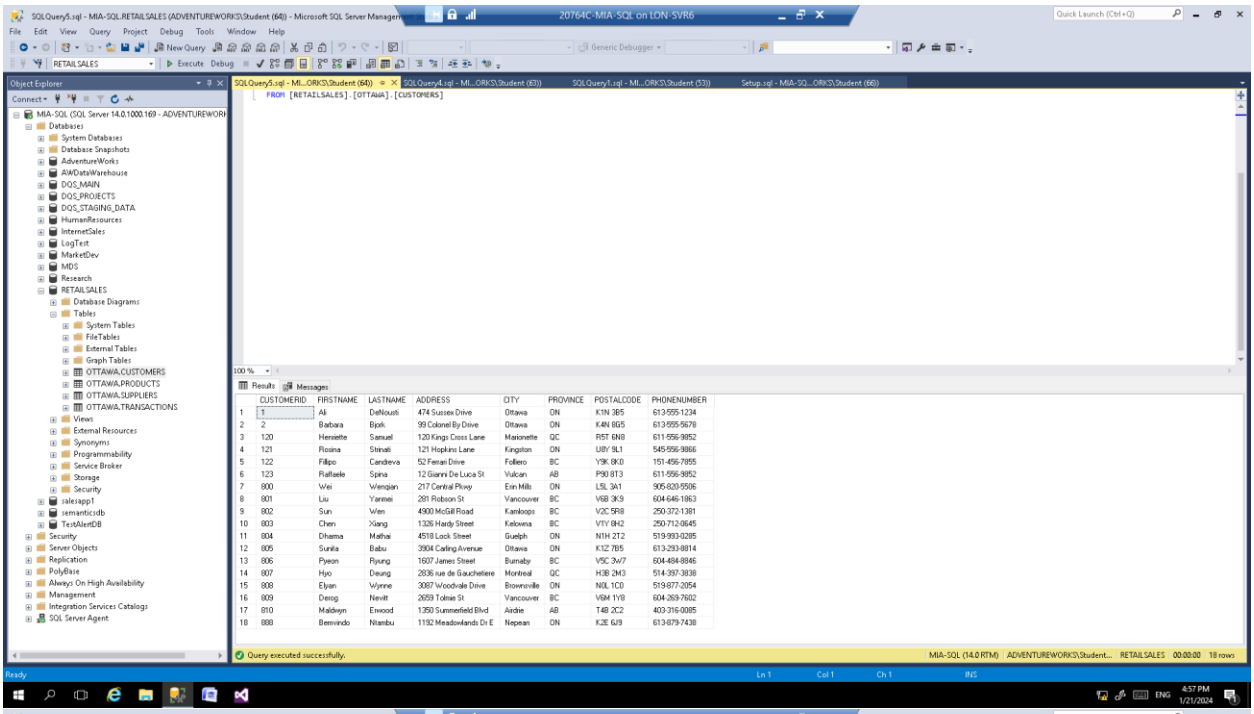
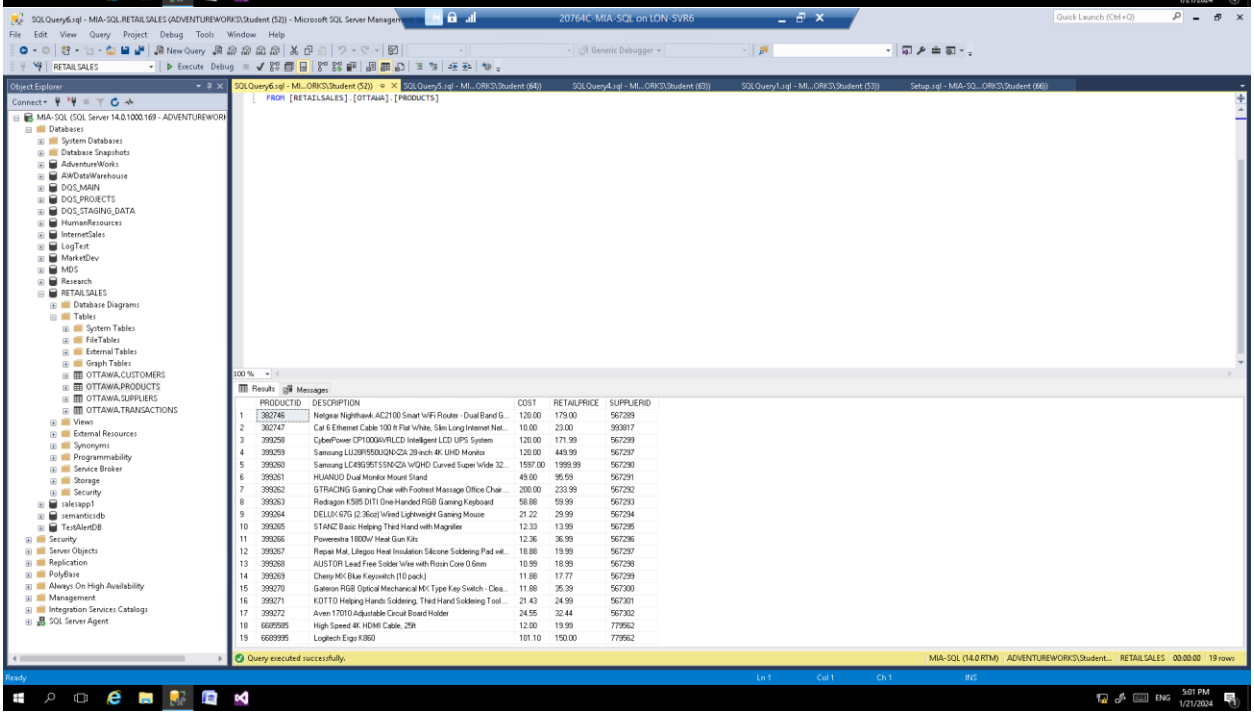
- Run a query that display the contents of all tables (do four separate queries)
    - Take a screenshot





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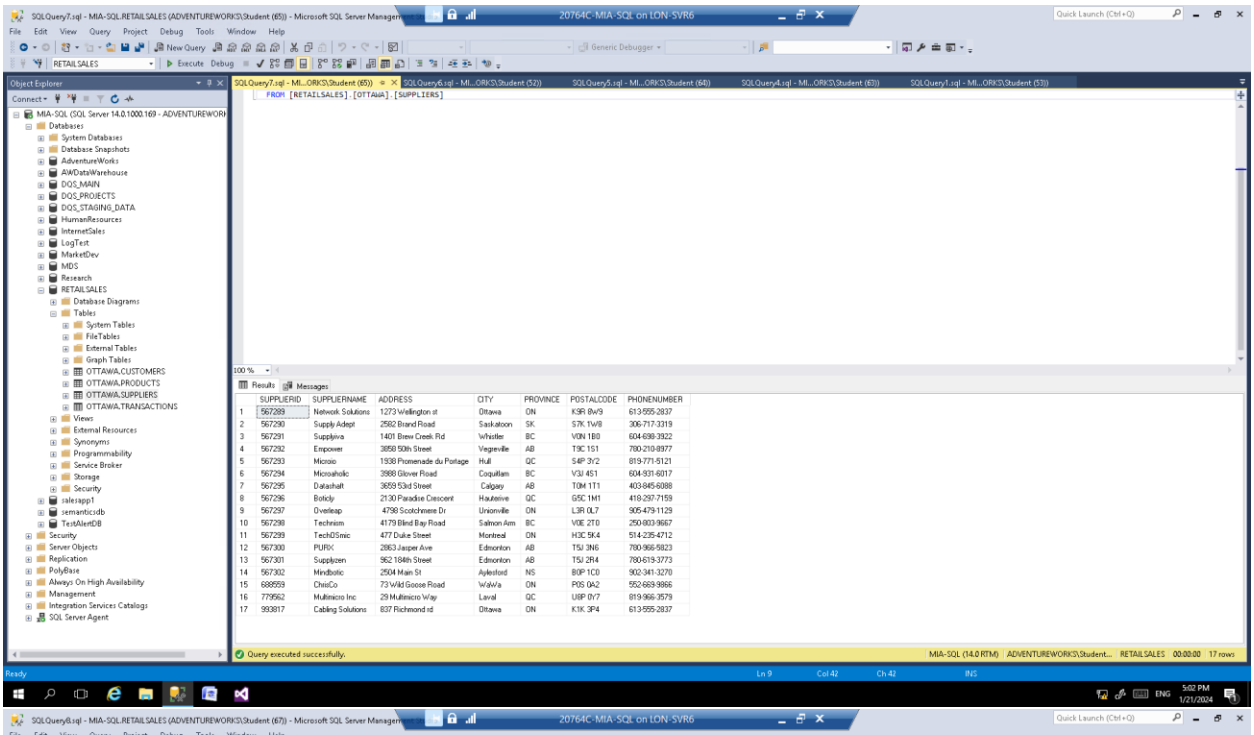

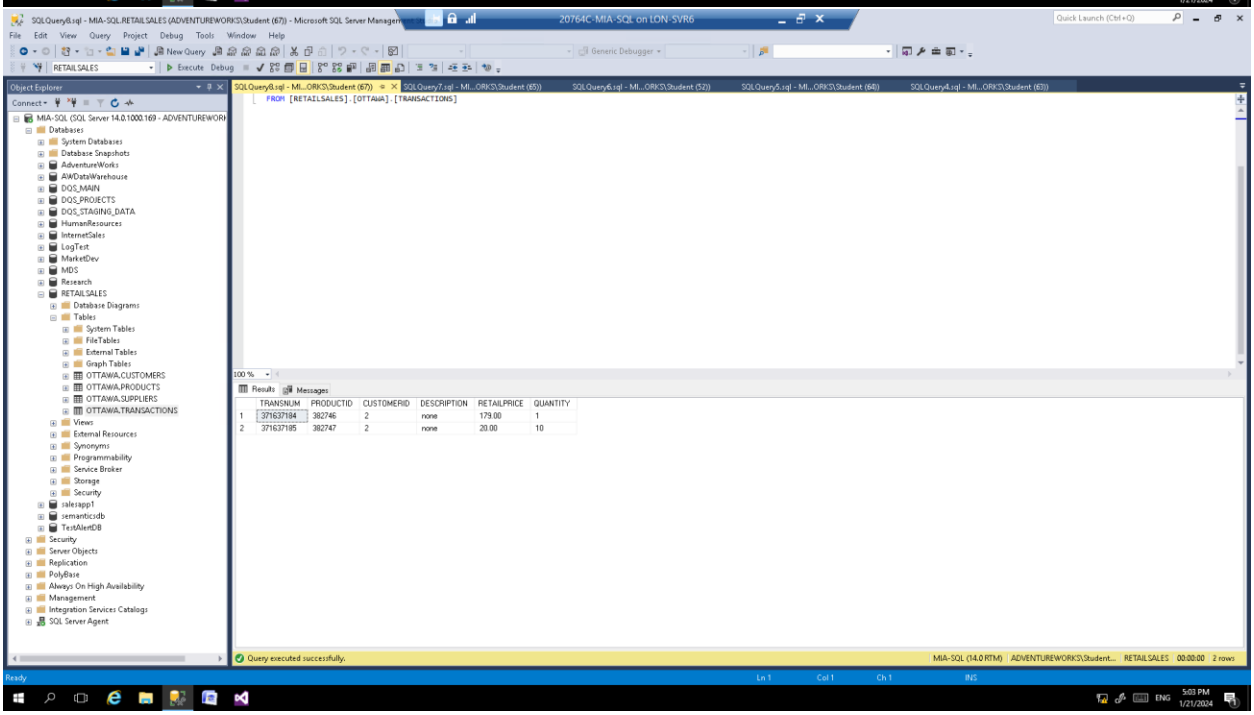

- Run a query that displays only the first 5 lines of the Customers table
    - Take a screenshot

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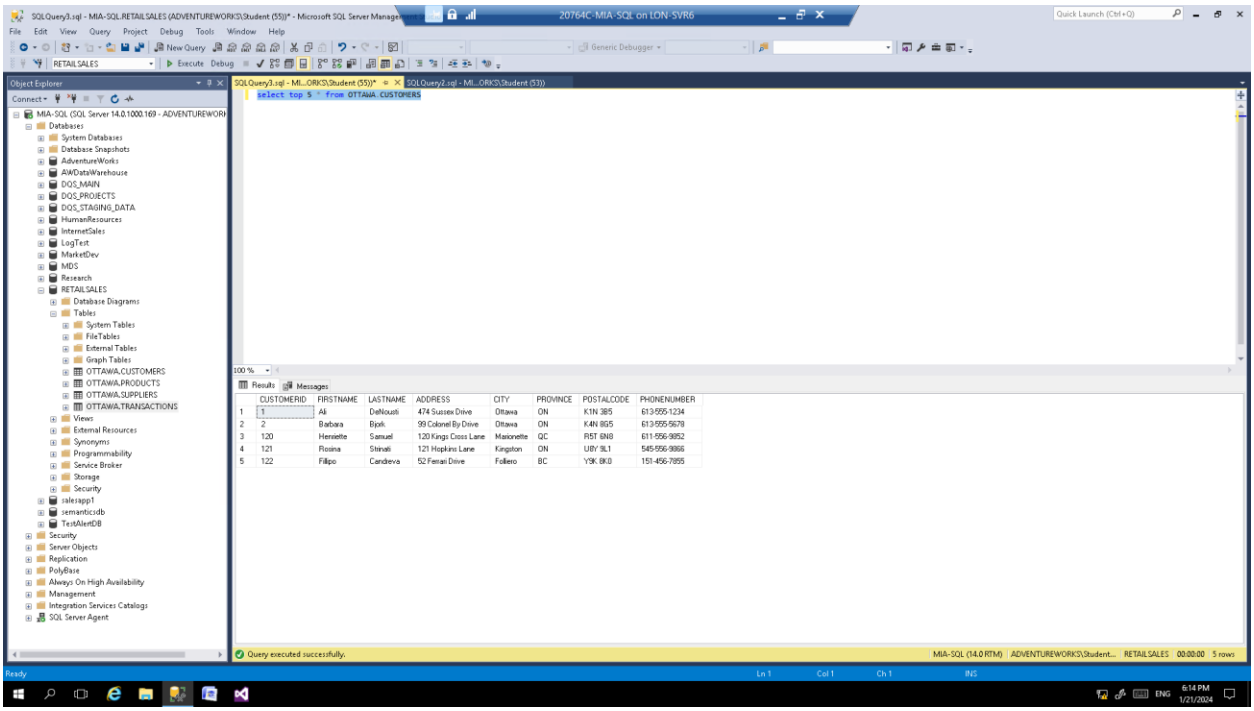
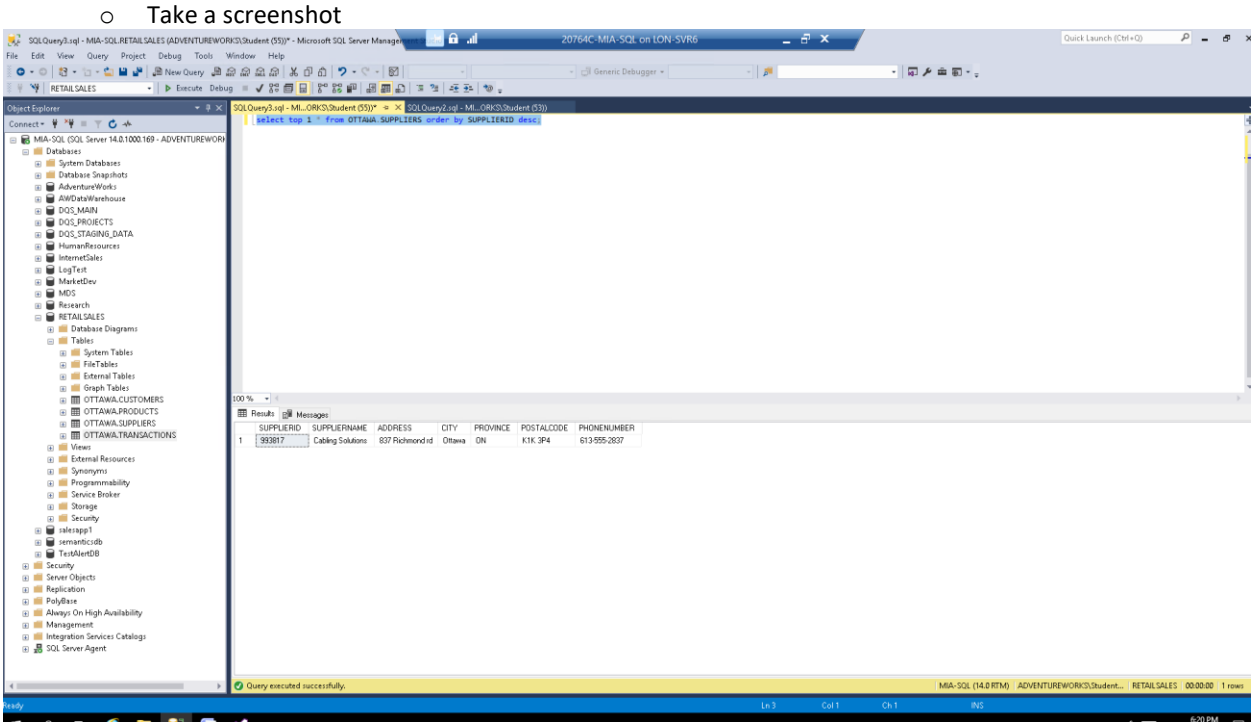


- Run a query that displays the last line in the Suppliers table
    - Take a screenshot



- Run a query that displays the names of the four tables that you created in the RetailSales database
    - Take a screenshot

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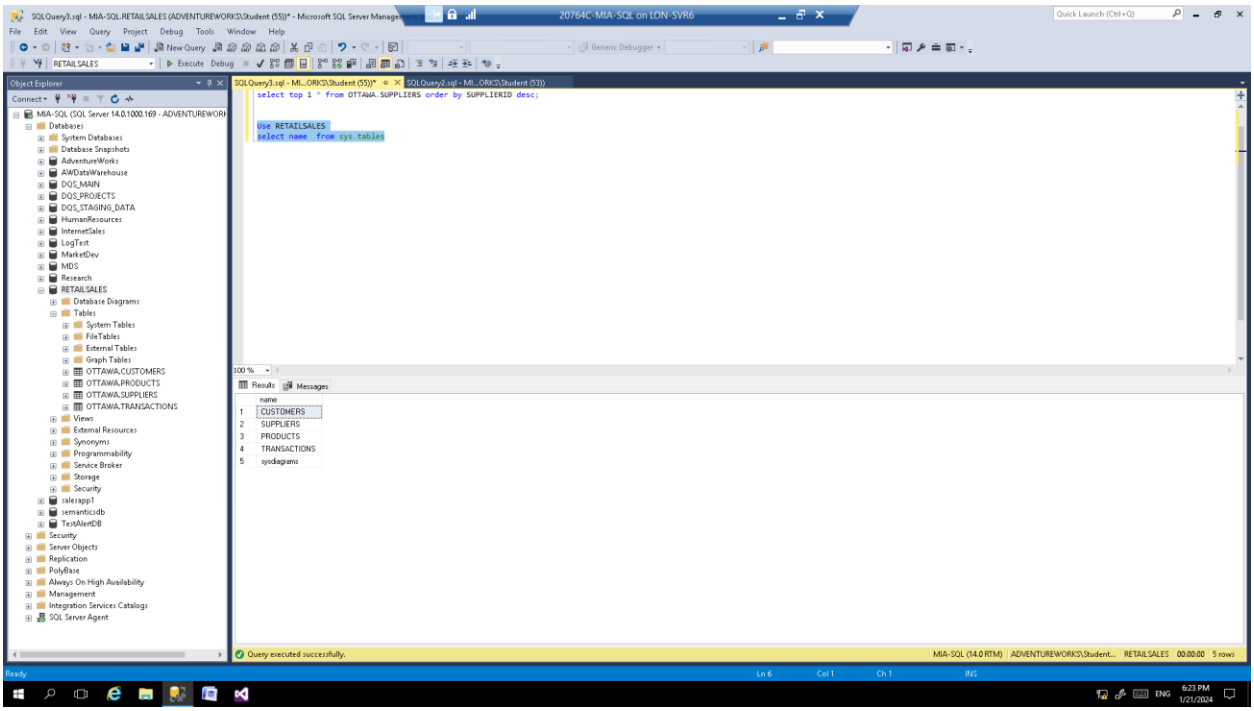

Backup

- Perform a Full backup of the RetailSales database
  - Take a screenshot that displays the size of the RetailSales database backup
  - Note: You can either run an SQL query on the size of the backup or take a screenshot of File Explorer showing the file size.

Table Information (Revisited)

Add the following information to the database into the appropriate table.

Customers Table	
CustomerID:	130
FirstName:	Tamara
LastName:	Egorova
Address:	6665 Kiseleva Way
City:	Seren
Province:	MN
PostalCode:	S9E 3R3
PhoneNumber:	955-556-2223
Suppliers Table	
SupplierID:	8859992
SupplierName:	Kalindi Corp
Address:	1 Starfield Way
City:	Cali
Province:	YK
PostalCode:	C8A L1E
PhoneNumber:	332-654-7855
Products Table	
ProductID:	62223335

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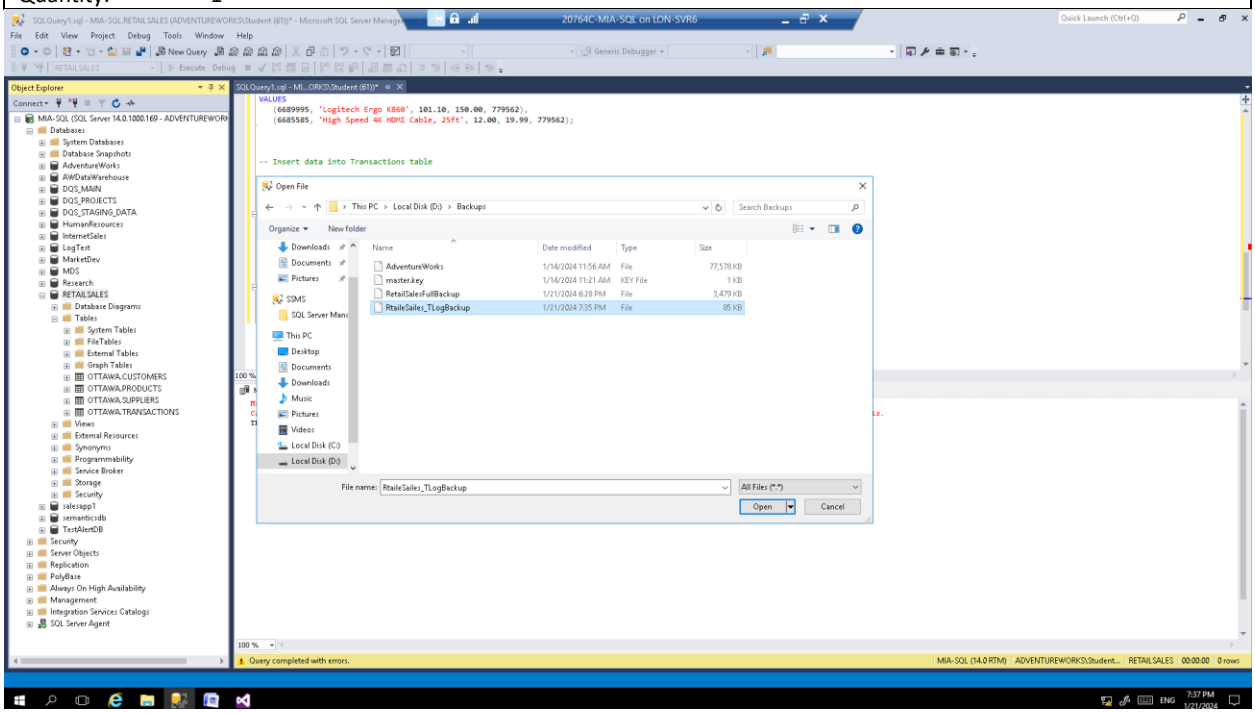
Description:	Projector Screen 100 Inch 16:9
Cost:	51.00
RetailPrice	109.96
SupplierID	8859992
Transactions Table	
TransNum:	55488855
CustomerID:	121
ProductID:	62223335
RetailPrice:	109.96
Quantity:	1



Backup (Revisited)

- Perform a transaction log backup of the RetailSales database
    - Take a screenshot showing the size of the transaction log backup

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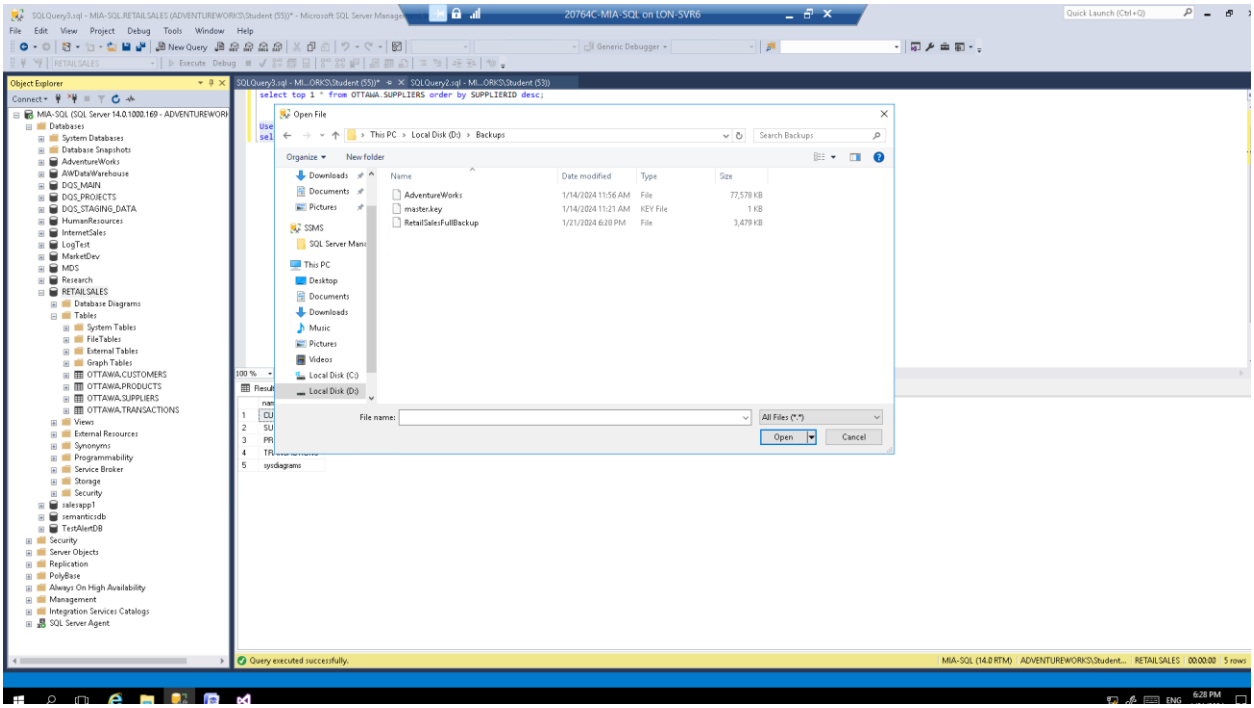
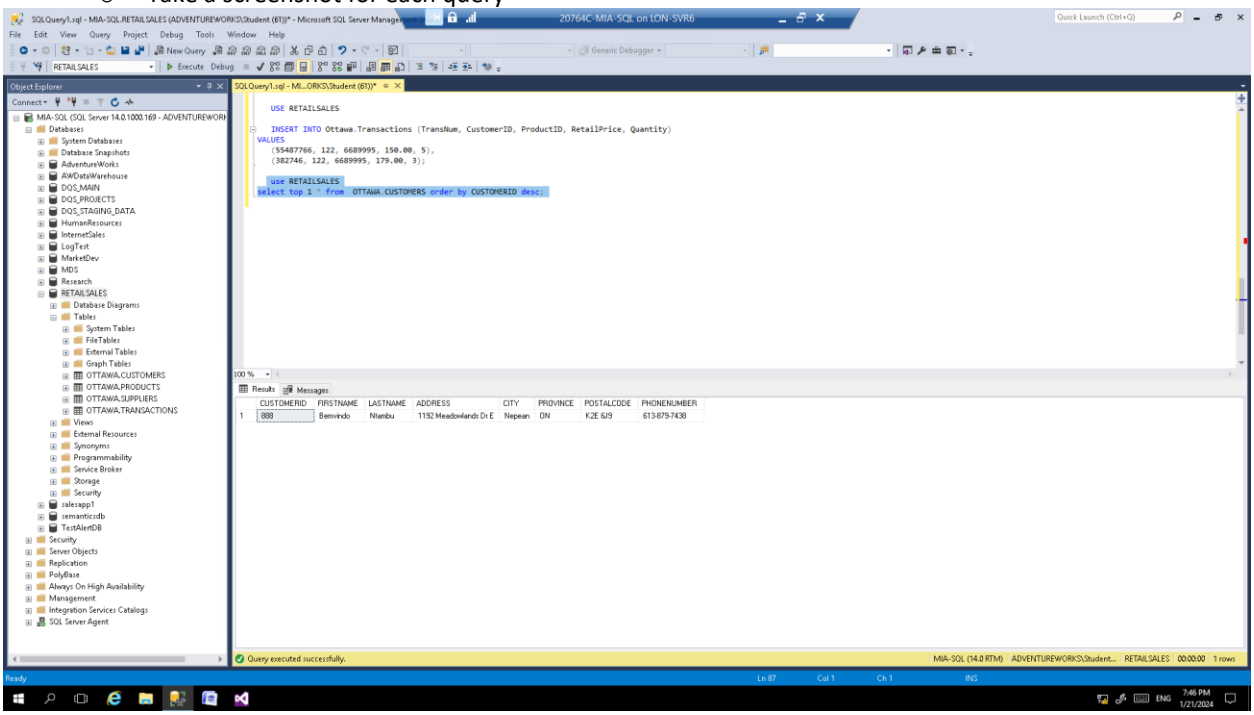


Query

- Run a query displaying the last line entered of each table
    - Take a screenshot for each query



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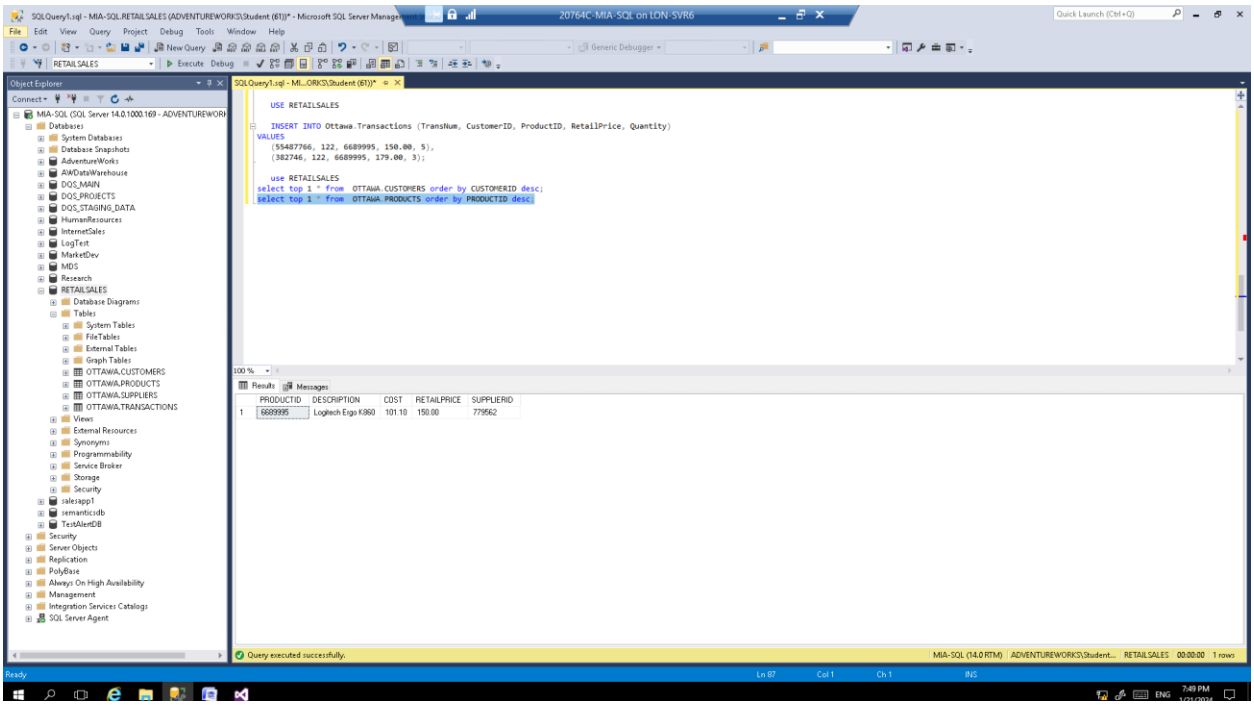

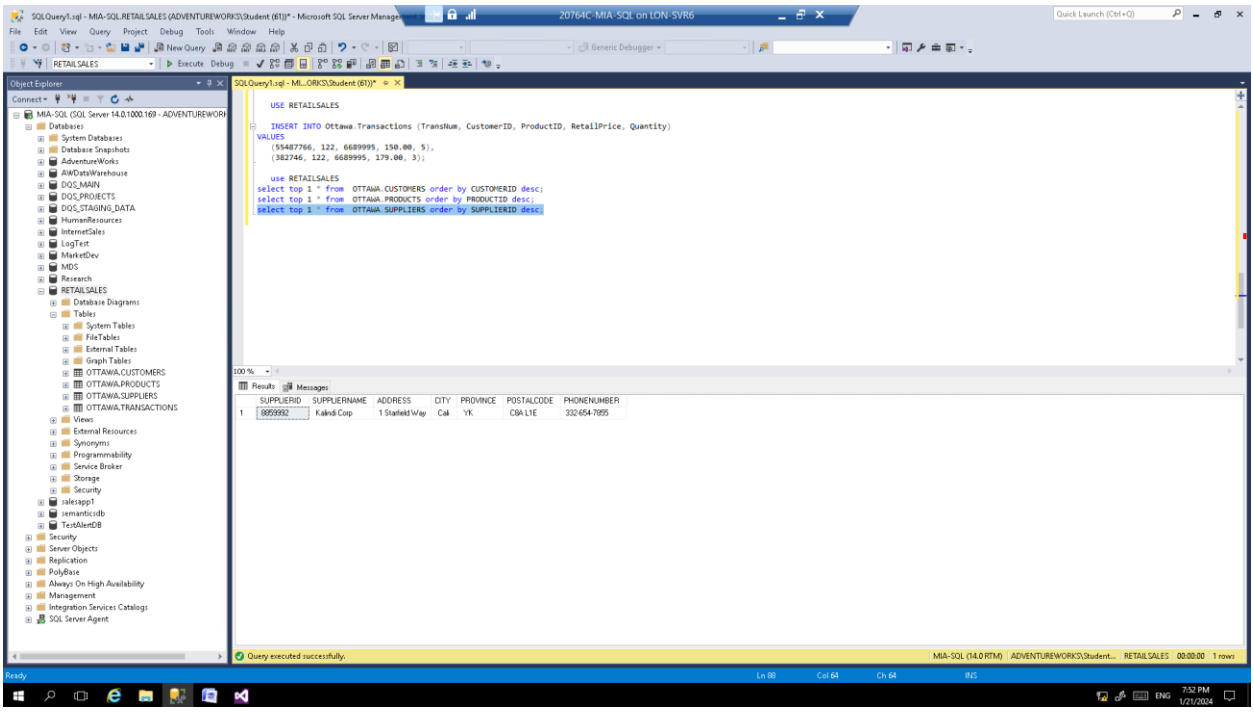

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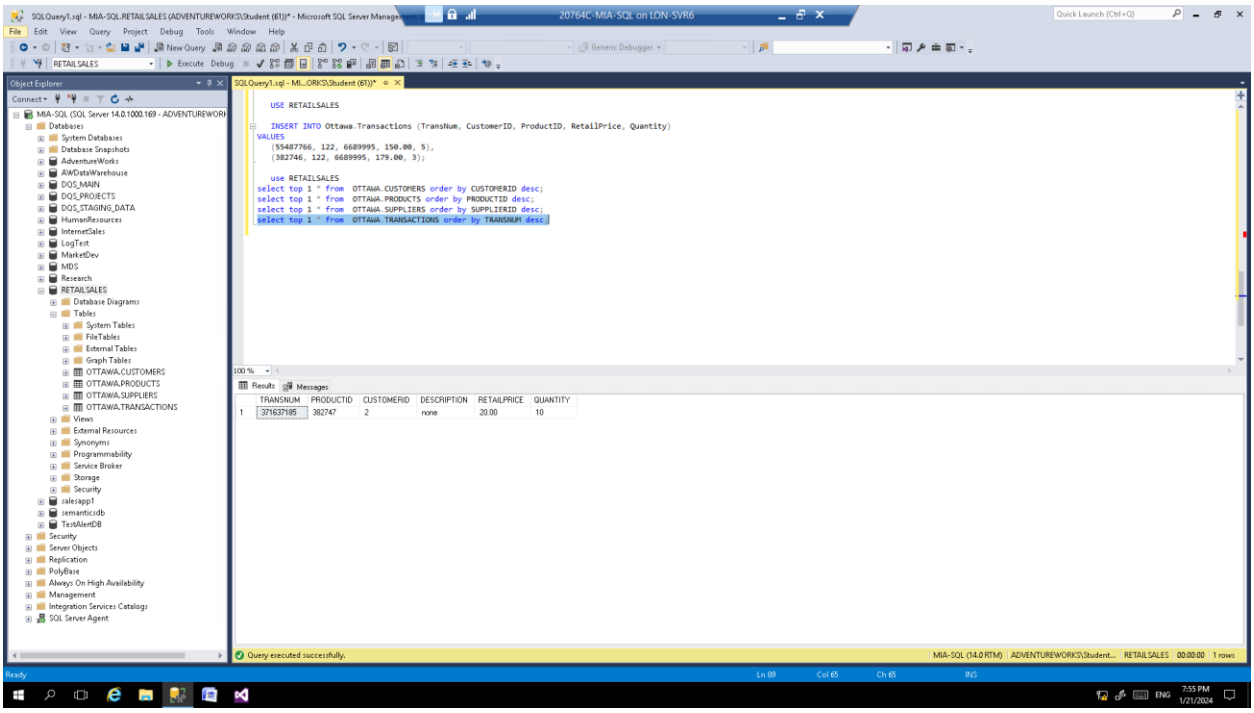

Security

- Create a **dynamic display mask** that hides the Customer Phone Numbers in the Customers table
- **Encrypt** the Cost field in the Products table
- Create a **dynamic display mask** that hides the address information in the Suppliers table

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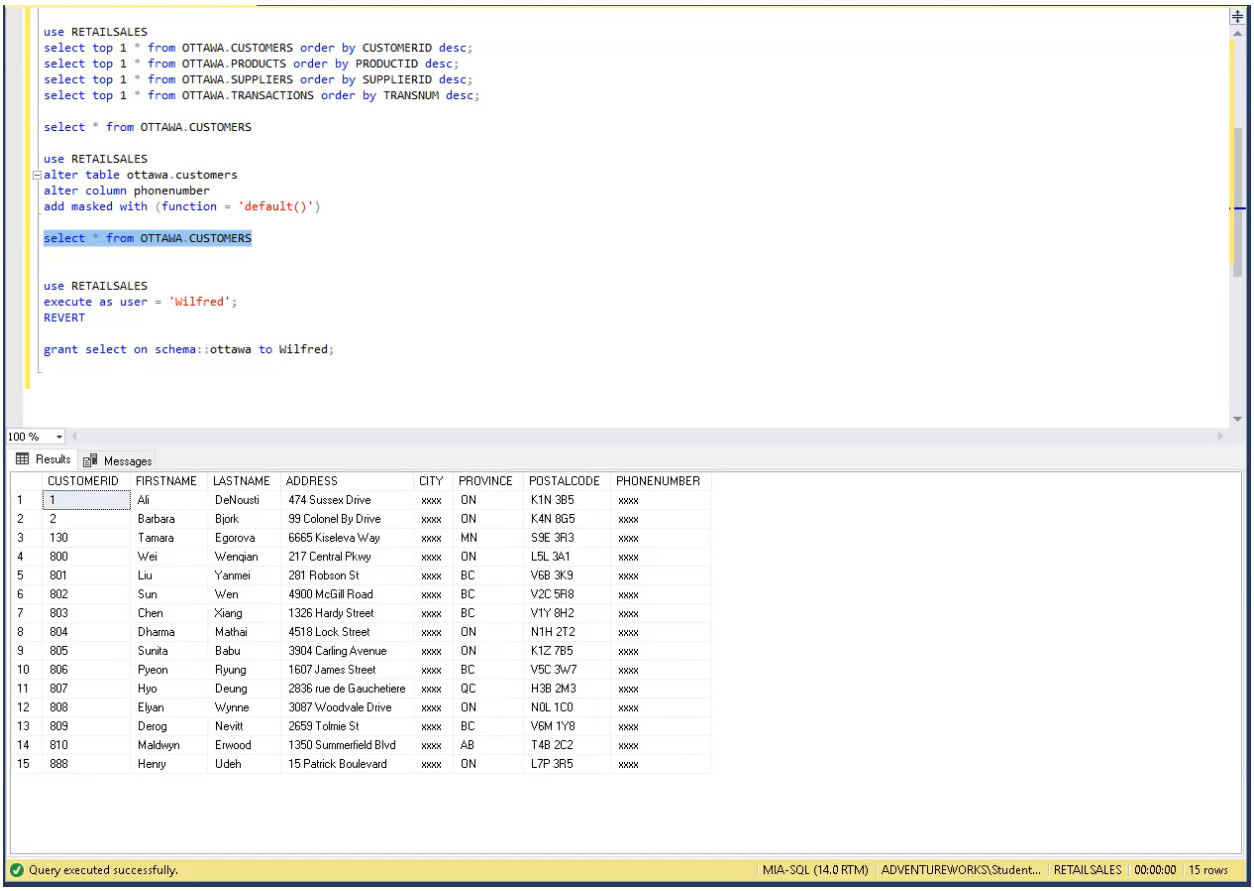

```
use RETAILSALES
select top 1 * from OTTAWA.CUSTOMERS order by CUSTOMERID desc;
select top 1 * from OTTAWA.PRODUCTS order by PRODUCTID desc;
select top 1 * from OTTAWA.SUPPLIERS order by SUPPLIERID desc;
select top 1 * from OTTAWA.TRANSACTIONS order by TRANSNUM desc;

select * from OTTAWA.CUSTOMERS

use RETAILSALES
alter table ottawa.customers
alter column phonenumber
add masked with (function = 'default()')

select * from OTTAWA.CUSTOMERS


use RETAILSALES
execute as user = 'Wilfred';
REVERT

grant select on schema::ottawa to Wilfred;
```

100 %

Results | Messages

	CUSTOMERID	FIRSTNAME	LASTNAME	ADDRESS	CITY	PROVINCE	POSTALCODE	PHONENUMBER
1	1	Ali	DeNousti	474 Sussex Drive	xxxx	ON	K1N 3B5	xxxx
2	2	Barbara	Bjork	99 Colonel By Drive	xxxx	ON	K4N 8G5	xxxx
3	130	Tamara	Egorova	6665 Kiseleva Way	xxxx	MN	S9E 3R3	xxxx
4	800	Wei	Wenqian	217 Central Pkwy	xxxx	ON	L5L 3A1	xxxx
5	801	Liu	Yanmei	281 Robson St	xxxx	BC	V6B 3K9	xxxx
6	802	Sun	Wen	4900 McGill Road	xxxx	BC	V2C 5R8	xxxx
7	803	Chen	Xiang	1326 Hardy Street	xxxx	BC	V1Y 8H2	xxxx
8	804	Dharma	Mathai	4518 Lock Street	xxxx	ON	N1H 2T2	xxxx
9	805	Sunita	Babu	3904 Carling Avenue	xxxx	ON	K1Z 7B5	xxxx
10	806	Pyeon	Ryung	1607 James Street	xxxx	BC	V5C 3W7	xxxx
11	807	Hyo	Deung	2836 rue de Gauchetiere	xxxx	QC	H3B 2M3	xxxx
12	808	Elyan	Wynne	3087 Woodvale Drive	xxxx	ON	N0L 1C0	xxxx
13	809	Derog	Nevitt	2659 Tolmie St	xxxx	BC	V6M 1Y8	xxxx
14	810	Maldwyn	Erwood	1350 Summerfield Blvd	xxxx	AB	T4B 2C2	xxxx
15	888	Henry	Udeh	15 Patrick Boulevard	xxxx	ON	L7P 3R5	xxxx

Query executed successfully. | MIA-SQL (14.0 RTM) | ADVENTUREWORKS\Student... | RETAILSALES | 00:00:00 | 15 rows

Query

- Run a query that shows the information in the Customers table has been masked
  - Take a screenshot
  - Note: Do not show me the unmasked data
- Run a query showing that the cost field in the Products table has been encrypted
  - Take a screenshot
- Run a query showing that the Suppliers address information is masked
  - Take a screenshot
  - Note: Do not show me the unmasked data

Submission

Screenshots of the successful completion of all your queries are required. You may paste the screenshots in the individual “Query” sections of this document.

You are also required to provide screenshots of the backup files that you created.

You must also include all the Transact-SQL syntax used.

```
INSERT INTO OTTAWA.CUSTOMERS
  (CUSTOMERID, FIRSTNAME, LASTNAME, ADDRESS, CITY, PROVINCE, POSTALCODE, PHONENUMBER)
    VALUES (
```

```
        130,'Tamara',' Egorova ','6665 Kiseleva Way ',' Seren ','MN',' S9E 3R3','955-556-2223'
        );

INSERT INTO OTTAWA.SUPPLIERS
   (SUPPLIERID,SUPPLIERNAME, ADDRESS, CITY, PROVINCE, POSTALCODE, PHONENUMBER)
     VALUES (8859992,' Kalindi Corp ','1 Starfield Way ',' Cali ',' YK ',' C8A L1E ','332-654-7855');



INSERT INTO OTTAWA.PRODUCTS
   (PRODUCTID, DESCRIPTION, COST, RETAILPRICE, SUPPLIERID)
     VALUES (62223335,' Projector Screen 100 Inch 16:9', 109.96,'8859992');
```